

Trade directories as a data source on occupational structure: evidence from England in 1851

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Abstract³

In the absence of occupational census data before 1851, recent research utilizes baptism registers to infer occupational structure in England in 1817. As an alternative source, we propose using trade directories as these seem to be unaffected by many of the fundamental issues stemming from baptism registers. We outline the history of trade directories and detail their construction. Using the 1851 occupational census and contemporary trade directories, we show that it is possible to infer local occupational structure from trade directories with reasonable accuracy. A suitably stratified sample could generate a national occupational distribution. This technique could likely be employed back to 1770.

Keywords: Census, structural change, industrialization.

JEL classifications: J21, N13, O14, Y10.

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0. Introduction. Recent characterizations of the British industrial revolution have played down the rate of economic growth, which is now widely agreed to have been slower than was suggested originally by Deane and Cole.⁴ Instead, more emphasis has been placed on the role of structural change, especially the transfer of labor resources from agriculture to industry.⁵ An obvious lacuna in this line of argument is that the available quantitative evidence on the rate or extent of structural change has been weak. The first census did not take place until 1801 and the occupation data that were collected in that year are worthless; households are categorized into three sectors (“Agriculture”, “Industry” or “Other”) and for most counties these sum to something like 50 per cent of the number of households, leaving us to wonder what the rest of the population were doing. Only with the census of 1841 do we get the first reliable estimates of occupational structure; but the 1841 categorization remains primitive and, in any case, by 1841 the first stage of industrialization was almost complete, so those data are not much help in measuring structural change. There have been previous efforts to quantify English social structure in the eighteenth century⁶; these have formed the basis of important quantitative research.⁷ But social structure is not exactly the same thing as occupational structure (even though the two are linked); and the quantification has been fairly broad brush and based on very imperfect sources.

Recent research by the team at Cambridge led by Shaw-Taylor and Wrigley has used a variety of sources – particularly baptism records and militia ballots – to address this issue. They seem to paint a very different picture to the one to which we are accustomed, with much higher rates of industrialization by 1801.⁸ We discuss their research much more fully in the next section. Overall, we believe that there are significant challenges with their approach and that it is essential to consider the evidence of alternative sources. Trade directories contain large amounts of contemporary data on economic and employment structure and offer a potentially fruitful line of enquiry. Trade directories are a well-known source for historians but there seems to have been no attempt to harvest them for large-scale quantitative analysis. The purpose of this paper is to assess the possibilities of such a line of research and consider the difficulties that it poses. We therefore undertake a detailed comparison of contemporary trade directory data and the 1851 census – the first census for which there are really detailed occupational data. We show that trade directories do seem to offer a reliable guide to occupational structure, when handled with suitable care. This could perhaps push our knowledge of occupational structure back from 1851 to 1770.

In the next section we consider the recent research. In section two we outline the history of trade directories and consider why and how they were compiled. In section three we undertake the comparison of trade directories and the 1851 census. Section four concludes.

⁴ Deane and Cole, *British economic growth*; Crafts and Harley, “Output growth”; Antràs and Voth, “Factor prices”.

⁵ Crafts, *British economic growth*.

⁶ Lindert and Williamson, “Revising”.

⁷ Crafts, *British economic growth*.

⁸ Kitson et al., “Creation of a ‘census’”; Shaw-Taylor *et al.*, “Occupational structure”; Wrigley, “PST system”.

1. Baptism records. From 1813 onwards, the registration of a baptism by the Church of England required the recording of the occupation of the father. For the period 1813-20, Kitson *et al.* took paternal occupation data from all extant registers (around 11 000) across England; for smaller parishes the population of registrations was entered, whereas sampling was used for larger parishes. These reported paternal occupations were used to infer occupational structure. Note that there are a number of sample selection issues here, some of which are discussed by Kitson *et al.*. Most obviously, the data pertain to males only, an important caveat that Kitson *et al.* state very clearly in their numerous papers.⁹ If males were employed systematically in different occupations – which seems highly likely – then the occupational distribution of males would not be a good guide to the overall occupational distribution. It is also problematic that many people were not Anglicans by this date – perhaps belonging instead to a dissenting Protestant church or the Catholic church – and thus will not appear in the baptism sample. Various occupations – such as the law, the military and the Church of England – were off limits to people who were not members of the Established Church; therefore Dissenters must have had a different occupational distribution. There are several, rather more subtle, sample selection issues that we discuss below.

In table 1 we present the occupational structure estimates of Kitson *et al.*, based on baptism registers, alongside those adopted by Crafts for 1801 and those available from the 1851 census. The Crafts data have been used repeatedly over the last 25 years as a basis for estimating economic growth; they show a very significant shift of labour resources out of the primary sector (agriculture, in fact) into the secondary sector. By contrast, the Kitson *et al.* data show a much higher share of industrial workers already by 1817, and a much lower share of service workers. This paints a very different picture of the rate of industrialization to that proposed by Crafts (and, later, Crafts and Harley).¹⁰ Kitson *et al.* find very little trace of industrialization in the early nineteenth century. Instead they find a Commercial Revolution, with a dramatic relative shift of employment out of agriculture and into services. Of course, their estimates are not strictly directly comparable to the other estimates in table 1 because they are based on male occupations only; some kind of correction would need to be made to generate a consistent series, although we are not in a position to say what the nature of that correction would be. Overall, we would say that either baptism registers must revolutionize the way that we think about British industrialization, or else we must have serious concerns about the validity of using them to infer occupational structure. We incline towards the latter point of view.

Table 1. Comparison of estimates of occupational structure.

	<i>1800</i> <i>(Crafts)</i>	<i>1817</i> <i>(Kitson et al.)</i>	<i>1851</i> <i>(Census)</i>
Primary	40	38	28
Secondary	30	42	41
Tertiary	30	19	32

Sources: 1800 – Crafts, *British industrialization*, 62; 1817 – Kitson *et al.*, “Occupational structure”, 10.

⁹ Kitson *et al.*, “Creation of a ‘census’”.

¹⁰ Crafts and Harley, “Output growth”.

Now we must ask what conditions Kitson *et al.* require for their inferences about occupational structure to be valid. There are two highly important ones.¹¹ First, it must be the case that representatives of all occupations were equally likely to marry and procreate. This is clearly false. The likelihood of marriage was obviously related to income – paupers, prisoners and the laboring poor (for example) would be less likely to marry than industrial workers. But the situation is more complicated than this because there are both lifestyle and lifecycle considerations. It seems plausible that males in some occupations would be systematically less likely to marry and procreate – such as merchant seamen who were away on long voyages, members of the Catholic clergy and faculty members of Oxford and Cambridge. Inferring the proportion of these men in the population on the basis of the frequency with which they appear in baptism registers seems hazardous because their lifestyle makes procreation less likely. Consider also the case of soldiers and Royal Navy personnel. Since they were abroad for years on end, one might conclude that they are in the same situation as merchant seaman. But this is rather a problem of *lifecycle* than of *lifestyle*. Most soldiers were young; most fathers were older. It may be the case that many of the males in the baptism register sample were *formerly* soldiers; but they are unlikely to be soldiers at the point in time when we observe them having children. Therefore occupations that were disproportionately filled by young people – who later moved on to another type of work – will be systematically under-represented in the baptism registers. Agricultural and other labourers occupies a significant share of the samples in Kitson *et al.* and thus represents an important source for errors. Note, in passing, that all the examples we have listed fall within the service sector – this will be of some importance in the following discussion.

There are also problems with multiple occupations. Many people had multiple occupations in the early nineteenth century and the question arises as to which occupation a father would offer if only one were recorded. If undertakers were commonly carpenters as well, then one could imagine that there may well be a systematic preference for reporting carpenter in preference to undertaker (or wagoner in preference to night soil collector, and so on). Kitson *et al.* discuss the assumption of equal marriage and fertility rates at some length and try to correct for the fact that it is violated, for example by adding extra laborers to their estimates. But their corrections are rather *ad hoc* and clearly extremely incomplete.

Second, Kitson *et al.* require that male completed fertility (that is, the total number of children produced per man) was the same for men across all occupations. We find this second condition – which Kitson *et al.* assume to be true – to be strong, surprising and unnecessary for the following reasons. It is a strong assumption in the sense that small violations have a large impact on the estimated occupational structure. Suppose that we sample four baptism records and find that two fathers are recorded as agriculturalists and two as industrial workers. Kitson *et al.* then infer that there were two agricultural workers and two industrial workers in the workforce (an industrial share of 50 per cent). Now suppose that each agricultural worker had one child and each industrial worker had two children in their lifetime. Then, of course, there would really be two agriculturalists but only one industrial worker in the workforce (an industrial share of 33 per cent). As we sketch in table 2 below, small variations in completed fertility across occupations could change Kitson *et al.*’s estimates of occupational structure drastically.

¹¹ As they note – Kitson *et al.*, “Occupational ‘census’”, 3-4.

Kitson *et al.* observe the data in column 2 of table 2 and assume a one-to-one mapping to the number of fathers, thus generating column 3 and therefore column 4. Now suppose instead that the completed fertility of service sector families (such as merchant seamen and soldiers) were one third lower than average, and the completed fertility of industrial families one third higher, as inscribed in column 5. Then the observations in column 2 map instead to the number of fathers in column 6, and then on to the occupational structure of column 7. But columns 7 and 4 in table 2 are simply a restatement of columns 2 and 3 in table 1. Thus differential fertility rates could easily transform the estimates of Kitson *et al.* into the estimates of Crafts. In reality, we do not need to make such an extreme assumption as a one third difference in sectoral fertility rates. If we admit that Crafts' estimates may be slightly out; and that there may have been some change already between 1800 and 1817; and if we allow for the fact that Kitson *et al.* consider only males; then a smaller assumed differential in fertility would be able to reconcile the two sets of numbers. But this need not lead to wholesale revisions of Crafts' position, nor of his description of the pattern of industrialization.

Table 2. Possible effect of variations in occupational fertility.

1	2 <i>Observed Baptisms</i>	3 <i>Inferred Number of Men (Kitson et al.)</i>	4 <i>Inferred Occupational Structure (Kitson et al.)</i>	5 <i>Hypothesized Occupation- Specific Fertility</i>	6 <i>Inferred Number of Men</i>	7 <i>Inferred Occupational Structure</i>
Primary	1.15	1.15	38	0.95	1.21	40
Secondary	1.25	1.25	42	1.35	0.93	30
Tertiary	0.60	0.60	20	0.65	0.92	30

Sources: see text.

We have now explained why we find the assumption of no variation in fertility across occupations to be strong. We find it also surprising because our understanding of the literature on English population growth is that a significant part of it was due to the movement of the workforce into industry. Changes in population growth between 1716 and 1816 were driven by changes in fertility; and changes in fertility were driven by changes in nuptiality, particularly the decline in the average marriage age of women by three years.¹² Earlier marriage led to higher completed fertility (more children per woman) because the number of births per women was largely determined by the number of years of marriage of fertile women. Industrial workers married earlier because they reached their peak earnings at a younger age; thus industrial workers had higher completed fertility than agricultural workers and the move into industry accelerated population growth. If this story were true then surely we should expect to see higher completed fertility for industrial workers in 1801 also? If we would accept this assumption, that would explain why Kits *et al* conclude with a secondary employment of 42% in 1817. This is higher than in 1851 (41%) and a significant higher level than Crafts estimation of 30% in 1801. In such circumstances, the contrary assumption – that completed fertility was the same across occupations – would not be innocuous.

We are also puzzled as to why Kitson *et al.* need to make this assumption at all. They entered all the available baptismal records. Since the name of the father is noted on

¹² Schofield, "British population change", 73-81.

the record, they could simply count the total number of children produced by each man in the sample. Since they know also his occupation, they could work out total fertility rates for each occupation. There are certain statistical complications involved in this exercise but – given the large size of their sample – they should be able to get precise estimates. This is an important issue to be resolved.

In general, Kitson *et al.* work at quite a high level of aggregation. That is, their results are presented for large geographical areas; occupations are highly aggregated, from 14 570 into only 113 occupations; and they pool data from many years. They note that some degree of aggregation is necessary because baptisms occur sufficiently infrequently that this generates small sample problems. This seems to be a significant drawback of using baptism records – there are not enough data to provide a really detailed picture.

Overall, we do not find any inexplicable inconsistencies between Crafts' data and those of Kitson *et al.*. Plausibly adjusting their data for differential fertility rates across occupations, the absence of women, and other sample selection issues, could reveal two estimates of occupational structure that are very similar. Since the two estimates anyway pertain to benchmark years that are 16 years apart, we certainly could not say that the two estimates are substantially different. Whether they are statistically significantly different is also impossible to say, since Kitson *et al.* do not provide confidence intervals for any of their estimates.

We stress that the primary purpose of this paper is not to provide a detailed critique of the research of the Cambridge team; we have not examined their sources and methods in sufficient detail to pass judgment. Rather, we seek to make the following points in our discussion of their research. First, the headline figures produced by Kitson *et al.* seem quite at variance with Crafts, whose figures were previously generally accepted. Second, the differential could be due largely to (unmeasured) differential fertility rates, the exclusion of women and sample selection. Third, an alternative data source exists that is abundant, not subject to these biases, and has not previously been employed – namely, trade directories. Therefore, it seems sensible to see if the trade directory data can be harnessed to generate reliable estimates of occupational structure. That is the purpose of this paper.

2. History of trade directories. Samuel Lee prepared the first British trade directory in 1677, but the entries covered only 1 953 wholesale merchants living in London.¹³ It seems to have met with limited success, since the exercise was not repeated until Henry Kent produced a new directory in 1734. Kent followed the same format as Lee but included 693 fewer names – so either London had shrunk or Kent's directory was very incomplete, the latter seeming more plausible. Coverage seems to have improved over the first few editions (up to 2 006 entries in 1740) but Kent's ambitions remained very limited in his subsequent annual revisions. Osborn's London directory first appeared in 1740 and offered a wider range of information, but was seemingly still very incomplete. The bar was raised in 1763 with the appearance of Mortimer's *Universal Directory*. He included not only the merchants and bankers of London but also people in other trades and professions: artists, musicians, doctors, lawyers, booksellers, shopkeepers and so on.

¹³ This paragraph is based on Goss, *London directories*, 1-35.

By the early nineteenth century, the Post Office directory, which first appeared in 1800, was listing around 11 000 entries; and Johnstone's 1817 directory was up to 27 000.

Importantly, Sketchley produced a directory for Birmingham in 1763 – the first for a town outside London.¹⁴ The first two editions of Sketchley's directory have not survived but the third edition (1767) has a format very similar to Mortimer's *Universal Directory* for London. Directories soon appeared for many other towns around England and thus trade directories could potentially constitute a very useful quantitative source on English economic history from 1763 onwards. Between 1763 and 1790, up to 50 new directories were produced. These covered ten towns and some also attempted to cover larger areas, with county directories appearing for Hampshire (1784) and Bedfordshire (1785). William Bailey, in 1784, was the first to attempt a national directory that covered the principal towns throughout the kingdom. Wilke's *Universal British Directory*, which appeared in eight volumes between 1791 and 1798, raised the bar again by including also many smaller towns.

In the early nineteenth century, town and county directories became common. In total, Norton's exhaustive survey counts 878 provincial (i.e. non-London) directories published before 1856. Many of these directories are readily available in electronic format because they are of interest to genealogists; therefore they constitute one of the most accessible historical sources. Over time, directories became more thorough and complete and were produced to a higher standard. Famous names – such as Pigot's and White's – started to appear in the 1810s; they set out to cover the whole country both systematically and repeatedly. From the perspective of the economic historian, repetition is a key ingredient. First, it may enable us to trace changes over time using a consistent source. Second, it probably generates a more accurate directory. How does repetition increase accuracy? The directory producer had an extra incentive to ensure that his directory was accurate because he had a reputation to maintain to generate future sales. He also had experience of producing directories and thereby a better idea of how to elicit accurate information (as we discuss further below). Finally, the directory producer already had local knowledge when preparing his directory (i.e. the data base generated by the previous edition).

The issue of accuracy is, of course, crucial in historical enquiry. First consider what we mean by accuracy. It is obviously not the case that the entire population was listed in a trade directory. Poor people would not have been listed; nor would many better off people who were not involved in trade (for example, retired people or military officers or noblemen). In fact, it is highly unlikely that even all the traders were recorded. There may be systematic omissions – such as dung collectors, who might not have wanted to advertise their trade – as well as random omissions and errors. In that sense, the directories are incomplete. But this does not make the directories useless. If we want to track accurately economic changes over time, or map variations across the country, then we do not necessarily need a complete register of all traders and producers. What we desire is transparency and, preferably, consistency. If we know the likely sources of error, so that we can correct for them; and if we know that these remained fairly constant over time; then we may be able to say something worthwhile about changes or variation in economic structure.

¹⁴ This paragraph is based on Norton, *Guide*, 1-15.

So how did directory producers compile their data? Several approaches seem to have been adopted.¹⁵ Early producers, such as Bailey and Pye, claim to have visited every house in the locality to elicit information from the householder. Pye, in fact, states that he gave up this approach in his later directories because it was too expensive, which sounds plausible. It may also have been counterproductive because people knocking unexpectedly at the door and asking about the nature of the householder's business might be suspected of being tax collectors – and therefore lied to, or told to go away. In any case, personal interview could not have been a practical mode of compiling county or national directories because the task was simply too vast for a private entrepreneur. Thus it became common to use local agents to collect information.

How could the directory producer ensure that the data he received from local agents were accurate? Wilkes partially solved this problem in the *Universal British Directory* by enlisting local printers and booksellers as his agents; since they were then remunerated in the form of offprints for local sale, they had a stake in generating an accurate product. Logically, the first thing that a potential purchaser would examine to gauge the accuracy of a national directory would be his own town: if it were accurate then he might be willing to believe that the rest of the directory were accurate; if it were not then it would be difficult for the local bookseller to persuade him otherwise. Thus each local bookseller was likely to be able to retail his free offprints of the national directory only if he did a good job of collecting the data in his own town.

Another innovative approach was crowdsourcing. A draft of the local directory was left with a prominent resident of the town and people were asked to inspect and correct it; perhaps this is where Jimmy Wales got the idea for Wikipedia. Traders and professional people had an obvious incentive to ensure that the information about them was accurate and up to date. Not only might this attract business from out of town but one could also imagine that there was a certain cachet derived from being in the directory. In the nineteenth century a class of professional directory agents evolved, who presumably varied in quality – and needed to be monitored – just as other professional workers did (and still do today).

The overall impression is that the quality and completeness of trade directories varied enormously. This probably depended on the integrity of the producer and may also have depended on the target audience. For example, Wilkes' *Universal British Directory* contains a short (or sometimes rather long!) description of the setting and history of each town. This would be particularly useful for someone travelling who wanted to plot an interesting route. These people were then likely to observe firsthand the quality of the directory because they would see how well the directory matched reality. By contrast, the Post Office *London Directory* was intended primarily for people sending letters, which might or might not arrive. A letter might go astray – or elicit no reply – for many reasons; or the postman might manage to deliver the letter even with a wrong address; so the user of the directory had no obvious way of assessing the accuracy of the directory.¹⁶

¹⁵ These are discussed in Norton, *Guide*, 16-18.

¹⁶ In the early nineteenth century the cost of postage was paid by the recipient, rather than the sender, so postmen were highly motivated to ensure that the letter reached its destination – even if the address were not accurate. Payment by the recipient obviously gave the postman the correct incentives to handle the mail carefully and promptly; otherwise postmen would have had an incentive simply to store bags of undelivered mail in their lofts, as they do today.

Thus considerable caution is warranted when using trade directories as a source. Consistency is probably more easily attained than accuracy. Depending on the historical issue being addressed, consistency may be a sufficient characteristic to ensure valid inference. Thus using the same directory for one place over time (such as multiple editions of Pigot's or White's), or using one directory that covered the whole country at a certain point in time (such as Wilke's *Universal British Directory*) are likely to be the two most trustworthy strategies. It also seems likely that earlier directories will be less accurate and complete than later directories – there was less competition to provide directories, a smaller database on which to build, less experience of eliciting accurate information and less infrastructure on which to rely (such as professional directory agents).

3. Testing trade directories against the census. The structure of the historical problem that we need to solve is sketched in table 3 below. We would like to be able to track occupational change over time using the census but there was no occupational census before 1851. We would therefore like to create a synthetic occupational census for earlier years using some other source. Since we have trade directories for 1770 and 1851 – and, indeed, at numerous intermediate dates – they are a potentially valuable source if we could harness them correctly.

Table 3. Data sources available to track occupational change.

1770	1851
?	Occupational census
Trade directory	Trade directory

Is it possible to move from trade directories to an occupational census with a sufficient degree of accuracy to make the exercise worthwhile? What are the difficulties that we face? The first problem is that trade directories tell us about the number of businesses operating in each occupation, not the number of workers employed. We will therefore need to multiply each business by an employment factor that is appropriate to that occupation. Note, however, that this procedure is strength as well as a weakness. When we multiply by employees per business, we implicitly include women as well men and those not affiliated to the Church of England.

The second problem is that the likelihood of a business appearing in the trade directory might be a function of its occupation. For example, it is plausible that businesses dealing directly with consumers (say, tailors) made sure that they were listed in the directory to obtain essential publicity, whereas businesses dealing with other businesses (say, ironworks) could successfully establish a reputation by word of mouth. If this were true then – even if we knew the average number of employees for each type of business – we would still not be able to estimate accurately the occupational structure of the population because we would have the wrong distribution of businesses across occupations.

We can lay these fears to rest using matched occupation and trade directory data from 1851. Logically, it should be possible to interpret the 1851 census as an enormous and complete trade directory for Great Britain. How? The 1851 census contains a table of

employees per business, broken down by occupation.¹⁷ Dividing the total number of people in each occupation by the average number of employees per business (in that occupation) should give the number of businesses in each occupation. That is, it forms a sort of national trade directory for Great Britain (albeit a trade directory with the street addresses and names of the businesses removed, which anyway are of no interest to us at this point). Census data are reported for each county and also major towns, as well as nationally; so it can be matched to town-level trade directories the we get a much finer geographical coverage than is possible with baptism records.

Of course, the procedure turns out to be rather more complicated than this. First, the 1851 table of employees per business enumerates only those businessmen (“Masters”) who have more than zero employees (“Journeymen and Apprentices”). So we have to infer how many businessmen there were who had zero employees. In principle, this is straightforward because, for each occupation, the table reports the number of employers having a particular number of workers. If we were to multiply all the employers in an occupation by the number of workers that each of them employed, then we should get the total number of people working in that occupation *except those businessmen who employed zero*. We could then compare this number to the total number of people recorded in the census as having that occupation. Any difference should (in theory) be composed of businessmen who had zero employees. The first problem with this exercise is that the number of employees is given only within certain bounds (1, 2, 3,... 10-19, 20-29,... 50- 74,... 75-100,... 350 and over). We address this problem by assuming that – on average – each firm was located mid-way between its particular set of bounds. For example, we assume that firms in the 10-19 category employed 15 workers; this is the most plausible assumption and – in expectation – will minimize the magnitude of any error.

The second problem is that most occupations have a very large discrepancy between the two estimates of total workers (i.e. the estimated number of workers employed is much lower than that enumerated in the census). This implies that many occupations had an implausibly high frequency of businessmen who employed zero workers. For example, in order to reconcile the two estimates of the number of people working as bakers, it would have to be the case that 75 per cent of bakers employed no workers. It is possible that 75 per cent of bakers employed no help, but it is not the most plausible suggestion. The census therefore seems to be internally inconsistent.

An explanation for such inconsistency is offered on p. cclxxvi of the 1851 census itself. Many employers neglected to complete the part of the form asking about the number of their employees. This would lead us to incorrectly assume that all the missing bakers (who were not recorded as employees) were sole proprietors with no employees. This would lead us to overestimate the total number of bakery *businesses* in Great Britain. For example, if a baker employed three people but neglected to note this in his census return then those three people would end up be counted as three one-man bakery businesses in our calculations. This could make it impossible for us to match the census with trade directories accurately.

We could therefore make one of two extreme assumptions. Either all the missing people in an occupation were one-man businesses; or all the businesses in that particular occupation employed people in the same size distribution that we observe in the table (i.e.

¹⁷ See British Government, *Census of Great Britain, 1851: Population Tables II*, vol. 1, cclxxvi-cclxxix.

for those firms that completed the form). This would be correct if some employers randomly neglected to complete that part of the census return. Logically, the truth will lie somewhere between these two extreme assumptions (i.e. there were actually some Masters who had zero employees and there some who neglected to fill in the form). We made all the calculations that follow using both of these alternative, extreme assumptions and found that it made no economically significant difference to our results. How can this be? It is because we are concerned only with the *distribution* of workers across occupations. If the employers in all trades were equally likely to ignore the part of the form dealing with the number of employees (for example, suppose that 50 per cent of all employers failed to complete it) then this will have very little effect on the estimated *distribution* of businesses.

If we make either of these assumptions, can we then accurately derive a national trade directory from the census? We cannot answer this question definitively without compiling all the data from a geographically complete set of 1851 British trade directories – a mammoth task that is far beyond this paper. But we can instead look at a sample of individual towns to shed some light on the issue. As well as giving the national and county data, the 1851 census reports the occupational structure of many English towns. Balancing our sample as far as possible in terms of size and geographical distribution, we entered the trade directory data for Whitehaven (Cumberland), Gateshead (Durham), Boston and Lincoln (Lincolnshire), Newark-on-Trent (Nottinghamshire), Kingston-upon-Hull (East Yorkshire) and Leeds (West Yorkshire).¹⁸ We made the calculations described above (based on each of the alternative assumptions) and then compared the total number of businesses estimated from the census to the total number of businesses recorded in the trade directories.¹⁹ The number of businesses recorded in the trade directories was much smaller, showing conclusively that the directories do not offer an exhaustive list of businesses in operation.

¹⁸ In the pdf of the census that is publicly available in the Chadwyck-Healey collection, data appear for only 34 towns. Most of these towns happen to be quite large and located in the north of England, viz: Chester, Macclesfield, Stockport, Carlisle, Whitehaven, Derby, Durham, Gateshead, South Shields, Sunderland, Blackburn, Bolton, Lancaster, Liverpool, Manchester and Salford, Oldham, Preston, Leicester, Boston, Lincoln, Newport, Newark-on-Trent, Nottingham, Newcastle upon Tyne, Tynemouth, Kendal, Kingston-upon-Hull, York, Bradford, Halifax, Huddersfield, Leeds, Sheffield and Wakefield. We began our work on the basis of these towns only and it is that which we report in this section, having matched the towns to contemporary trade directories as far as possible. We later discovered, by going back to the printed copy of the census, that data are provided for many other towns – but these were erroneously missed out of the pdf file that is publicly available from Chadwyck-Healey. To make our sample more complete, we later added “Greater Birmingham” (that is, Birmingham, Bromesgrove, Burton-on-Trent, Cheadle, Droitwich, Dudley, Evesham, Kidderminster, Leek, Litchfield, Newcastle-under-Lyne, Penkridge, Pensnall, Pershore, Stafford, Stoke-on-Trent, Stone, Stourbridge, Tamworth, Tenbury, Upton-on-Severn, Uttoxeter, Walsall, West Bromwich, Wolverhampton, Worcester).

¹⁹ A small number of occupational terms used in the census were not used in the trade directory. For example, no business is listed as a “Fustian manufacturer”; since fustian was a type of fine cotton cloth, those businesses were presumably listed as “Cotton manufacturer”. The same is true of “Thread manufacturer” and “Calico and cotton printer”. We therefore aggregated workers in those industries (as reported in the 1851 census) with cotton manufacturers and calculated one multiplier for all branches of the cotton industry than we applied to each of its components (cotton, fustian, thread and printing). For “Weaver (material not stated)” we took the multiplier to be the average of cotton, flax and woollen manufacturers. For “Skinner” we took the multiplier to be the average of other occupations in the sub-class (which were all very similar); and the same for “Fuller”.

But recall that we are not actually trying to find the *number* of businesses. All we are trying to discover is the *distribution* of businesses (and, from there, the distribution of individuals' occupations). Were the distributions of businesses across occupations the same in the census and the trade directories? Yes. How can we summarize their similarity in some type of descriptive statistic? Calculate the percentage of total businesses constituted by each occupation in both the census and the trade directory. That is, work out what percentage of businesses were bakeries, tailors, taverns, and so on. Now regress the trade directory distribution on the census distribution. What should you expect to find if the trade directory is a random sample of businesses in a particular town? Then a one per cent larger share accruing to a particular occupation in the census will be reflected by a one per cent larger share accruing to that occupation in the trade directory (i.e. the coefficient on the census data will be unity). So if bakeries and taverns comprised five per cent and ten per cent respectively of the population of businesses in a town, according to the census, then they should similarly comprise five per cent and ten per cent respectively of the businesses recorded in the trade directory.

Of course, to the extent that there is measurement error in the estimated occupational structure derived from the trade directory, the estimated coefficient in the regression will be biased downwards, for standard econometric reasons. Hence we expect to observe estimated coefficients that are less than unity but hopefully not statistically significantly different from it. If the overall distributions are quite similar then the fit of the regression (the r-squared) will also be high. Note that some of the trade directories that we matched against the 1851 census were compiled several years after the census; we chose them simply because they were the closest years available. Such temporal mismatch would be expected to induce more measurement error and bias the results towards rejecting the hypothesis that the trade directories and the census exhibit the same occupational distribution. Note further that this need not generally be a problem with using trade directories. We are constrained here to find trade directories as close as possible to 1851 because we are undertaking a direct test against the census. If we were given a free choice of year, and were simply trying to assemble a set of trade directories that gave a good coverage, then there would be less temporal mismatch. We would simply construct an occupational distribution for whichever year had the best trade directory coverage (such as 1770 or 1784, when Bailey's first national directory appeared).

We undertook the regression exercise for our sample of towns and found that the distributions of the census and trade directories were fairly similar for each town, and the coefficient on the census was not significantly different from unity. We report these regressions in table 4 below. These results suggest that the 1851 census can generate an occupational distribution of businesses that mirrors that found in trade directories – both at the local and national levels. The results also imply that it is safe to work in the other direction – i.e., infer the occupational distribution that we would observe in the census from contemporary trade directories.

Table 4. Regressing trade directory occupational shares on those of the census, c. 1851.

	<i>Coefficient</i>	<i>95% confidence interval</i>	<i>r²</i>	<i>N</i>
Greater Birmingham	0.86	0.75 – 0.97	0.71	97
Boston	0.95	0.79 – 1.10	0.70	64
Gateshead	0.91	0.75 – 1.08	0.66	61

Kingston upon Hull	0.85	0.70 – 1.00	0.65	70
Leeds	0.92	0.82 – 1.03	0.79	82
Lincoln	1.01	0.86 – 1.15	0.73	72
Newark	1.00	0.83 – 1.16	0.71	60
Whitehaven	0.93	0.75 – 1.12	0.57	76
Pooled sample	0.99	0.90 – 1.09	0.78	119

Notes. We exclude all occupations for which there are zero workers and all occupations for which this is no multiplier available from the table of employees per business. We aggregated “Builders” with “Mason (pavior)” and “Bricklayer”; we excluded “Merchants” because the multiplier in the table of employees per business is based on only three observations in the entire country; and we excluded the top five and bottom five occupations (in terms of their distance from the occupational share reported in the census) in each town. Our rationale for the last step was that there were a small number of very large outliers that were drastically and randomly skewing the results, and most of these outliers were obviously problematic. For example, “Coal miners” seem to be massively underreported in the trade directories, compared to the census. But this is easily understood when we see that the table of employees per business reports an average of 49 miners per coal mine, which must surely be a drastic underestimate. In general, it was more or less the same 10 occupations that were problematic in each of the towns (notably, “Straw hat and bonnet maker”, “Woollen cloth manufacture”, “Flax, linen manufacture”, “Coal merchant, dealer”, “Shopkeeper (branch undefined)” and “Hosier, haberdasher”). The number of observations differs for each regression simply because some towns have more occupations than others.

We feel that these results are satisfactory and act as a “proof of principle”: occupational structure can be inferred from trade directories. However, the approach outlined above is not an optimal solution for a number of reasons. First, the table of employees per business in the 1851 census is truncated: the largest size bracket in the table is for those employing “350 or more”. Thus those establishments employing 350 are lumped in with those employing several thousand. Since some industries, such as cotton, are likely to have had systematically larger establishments than other industries, this could well introduce a bias into the results. Second, some industries provided very few returns; for example, only three merchants in the whole of Great Britain reported the number of people that they employed. So the figure for employment by merchants is much less reliable than the figures for industries in which thousands of returns were received.

Third, relying on the employment table in the census throws away important information. We have fitted the trade directory data to the census data using the employment table and shown that this gives coherent results. But is this the best that can be done? No. The employment table is very imperfect. And we know exactly how imperfect it must be. If we divide the census data by the trade directory data then we can create our own employment table. This table is exact, in the sense that it matches the two data series perfectly, by construction. Since this is the best that we can hope to do, it is logical to use this inferred table of employment in place of the one found in the census, even though the census table performs adequately. Fourth, and perhaps most importantly, the census table of employees per business does not cover all the occupations recorded in the census. In fact, it covers only around 240 of them, out of 369 in total. Some individual occupations are retained (such as “Iron founder”); but many of them are aggregated into broader categories (such as “Other iron workers”). This is hugely disappointing because we would like to study in detail the changing pattern of industrial production. If most of the individual occupations are aggregated then the coarseness of the resulting occupational structure will preclude us from being able to offer a precise description of England’s industrialization. The trade directories record businesses in

several thousand distinct occupations, so if we had an employment table that covered all the 369 occupations reported at the town level in the 1851 census then we could aggregate the trade directory data in such a way as to produce a national occupational census for earlier years that was exactly analogous to the census tables of 1851. Moreover, this set of 369 occupations was designed by the Registrar General in 1851 to encompass all possible occupations in the economy, so it is coherent and complete. For these reasons, we believe that the constructed table of employees per business offers a superior solution to using that found in the 1851 census.

An obvious problem with using the trade directories predating 1851 is that there is no table of employees per business before that produced in the 1851 census. One must be created, and two obvious strategies commend themselves. First, this could be done from contemporary historical sources. Take contemporary accounts of cotton factories, ironworks, coal mines, bakeries and so on; note, on average, how many employees there were per business. This could be done for each date; average employment per business (occupation) could be allowed to change over time. If the historical record on business size were sufficiently rich, then this would be the most precise approach. Second, we could impose on earlier years the employees per business table from 1851. This has the disadvantage that it is temporally removed, but the advantage that it is accurately observed; we are trading off two types of error.

Suppose that we construct a table of employees per business by dividing the census population by the number of businesses in the trade directory. This actually conflates two effects. First, there are a certain number of employees per business. Second, there is under-registration of businesses. Suppose that there were actually two employees per baker but only half of the bakers appeared in the trade directory. Then we will infer erroneously that there were four employees per baker. Is this a problem? Not necessarily, for the following reason.

We would use the table of employees per business for 1851 to reflate our register of businesses in (say) 1770. This generates a synthetic occupational census for 1770. The only thing that is important for this method to be valid is that the weights reflected in the table are stable between 1770 and 1851. This will occur most obviously if all the components are stable (there are always two employees per baker and bakers always appear in the trade directory with a 50 per cent probability). But the requirements for our table to be functional are actually much weaker than this. For example, suppose that all businesses have a 50 per cent probability of appearing in the trade directory in 1851, but only a 25 per cent probability in 1770. This would not bias our results because the estimated employment in all occupations would be falling proportionately to one another – so our estimate of the distribution of workers across occupations would be unaffected. Suppose that establishment size was rising in all occupations: there were two bakers per bakery in 1770 but four bakers per bakery in 1851. This would not generate any bias as long as establishment size was rising at the same rate in all occupations.

By contrast, it is highly likely that differential changes across occupations in the frequency of business registration in the trade directory, or differential changes in establishment size, would reduce the accuracy of our estimated distribution of the working population in 1770. That is, unless changes in frequency and changes in establishment size happened to offset each other. We have no way of knowing whether there were differential changes in the table of employees per business between 1770 and

1851 without constructing a table of employees per business in 1770, relying on contemporary sources. However, if we opted to rely solely on the 1851 table then we could at least consider the likely direction of any such changes in business size (such as the increase in the size of cotton mills) and ask to what extent – and in what direction – our estimated occupational structure might be biased. This type of sensitivity analysis can help us to put plausible bounds on the scale of the problem.

4. Conclusions. Trade directories are numerous and easily accessible for the period 1770 to 1856. But, surprisingly, they have not previously been exploited in a systematic way. Although the quality is variable, the producers of trade directories employed several innovative and wily techniques to elicit complete and accurate information. For example, they used crowdsourcing and local booksellers to collect information, paying the latter in free offprints (which would only have a high retail value if the directory were of good quality). The trade directories constitute a potentially valuable untapped source of information on occupational structure at a crucial stage of Britain's industrialization. Importantly, trade directories do not suffer from the gender, religious, lifestyle and lifecycle biases inherent in other sources, such as baptism records.

We have shown that trade directories can be used to generate an employment distribution across occupations that mirrors the true distribution, as revealed by the census. We tested this contention for the year 1851 – the first year for which we have a detailed occupational census. Regression analysis shows that a one percent increase in the share of an occupation in the trade directory is, on average, matched by a one percent increase in the share in the census (controlling for the number of employees per business). There are alternative procedures for controlling for the number of employees per business, such as using contemporary sources or the data from 1851; each approach has its advantages and disadvantages.

A worthwhile next step would be to undertake a large-scale exercise for an earlier benchmark year – such as 1784 or the mid-1790s, when we have nationwide trade directories – to gauge the results of a rigorous and complete analysis. It is to be hoped that such an attempt will be completed in the near future.

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Appendix 1. Town-level occupational breakdown of the 1851 census.

ID	Class	Sub-class	Occupation
1	1	1	Post Office
2	1	1	Inland Revenue
3	1	1	Customs Service
4	1	1	Other Government officers
5	1	2	Police
6	1	2	Union relieving officer
7	1	2	Officer of local board
8	1	2	County, local, -officer (not otherwise distinguished)
9	1	3	East India Service
10	2	1	Army officer
11	2	1	Army half-pay officer
12	2	1	Soldier
13	2	1	Chelsea pensioner
14	2	2	Navy officer
15	2	2	Navy half-pay officer
16	2	2	Seaman, R. N.
17	2	2	Greenwich pensioner
18	2	2	Marine
19	2	2	Others engaged in defence
20	3	1	Clergyman
21	3	1	Protestant minister (not otherwise described)
22	3	1	Priest of other religious bodies
23	3	2	Barrister, advocate, special pleader, conveyancer
24	3	2	Solicitor, attorney, writer to signet
25	3	2	Other lawyers
26	3	3	Physician
27	3	3	Surgeon, apothecary
28	3	3	Other medical men
29	3	4	Parish clerk, clerk to church
30	3	4	Other Union, district and parish officer
31	3	5	Law clerk
32	3	5	Law stationer
33	3	6	Druggist
34	3	6	Others dealing in drugs
35	4	1	Author
36	4	1	Editor, writer
37	4	1	Others engaged in literature
38	4	2	Painter (artist)
39	4	2	Architect
40	4	2	Others engaged in the fine arts
41	4	3	Scientific person, observatory and museum keeper
42	4	4	Music-master
43	4	4	Schoolmaster, schoolmistress
44	4	4	Governess
45	4	4	Other teachers
46	5	1	Wife (of no specified occupation)
47	5	2	Widow (of no specified occupation)
48	5	3	Son, grandson, brother, nephew (not otherwise returned)
49	5	3	Daughter, granddaughter, sister, neice (not otherwise returned)
50	5	4	Scholar – under tuition at home
51	5	4	Scholar – under tuition at school or college
52	6	1	Innkeeper
53	6	1	Innkeeper's wife
54	6	1	Lodging-house keeper
55	6	1	Officer of charitable institution
56	6	1	Others – boarding and lodging
57	6	2	Domestic servant (general)
58	6	2	Housekeeper
59	6	2	Cook
60	6	2	Housemaid
61	6	2	Nurse

62	6	2	Inn servant
63	6	2	Nurse at hospital, etc.
64	6	2	Midwife
65	6	2	Charwoman
66	6	2	Coachman
67	6	2	Groom
68	6	2	Gardener (servant)
69	6	3	Hairdresser, wig-maker
70	6	3	Hatter
71	6	3	Straw hat, bonnet, -maker
72	6	3	Furrier
73	6	3	Tailor
74	6	3	Cap, -maker, dealer
75	6	3	Milliner, dressmaker
76	6	3	Shirtmaker, seamster
77	6	3	Shawl manufacturer
78	6	3	Staymaker
79	6	3	Hosier, haberdasher
80	6	3	Hose (stocking) manufacture
81	6	3	Laundry-keeper, mangler
82	6	3	Rag, -gatherer, dealer
83	6	3	Glover (material not stated)
84	6	3	Shoemaker, bootmaker
85	6	3	Shoemaker's wife
86	6	3	Patten, clog, -maker
87	6	3	Umbrella, parasol, stick, -maker
88	6	3	Others providing dress
89	7	1	House proprietor
90	7	1	Merchant
91	7	1	Banker
92	7	1	Ship-agent
93	7	1	Broker
94	7	1	Agent, factor
95	7	1	Salesman, saleswoman
96	7	1	Auctioneer, appraiser, valuer
97	7	1	Accountant
98	7	1	Commercial clerk
99	7	1	Commercial traveller
100	7	1	Pawnbroker
101	7	1	Shopkeeper (branch undefined)
102	7	1	Shopkeeper's wife
103	7	1	Hawker, pedlar
104	7	1	Other general merchants, dealers, agents
105	8	1	Railway engine, -driver, stoker
106	8	1	Others engaged in railway traffic
107	8	2	Toll collector
108	8	2	Coach and cab owner
109	8	2	Livery-stable keeper
110	8	2	Coachman (not domestic servant)
111	8	2	Carman, Carrier, carter, drayman
112	8	2	Omnibus, -owner, conductor
113	8	2	Others engaged in road conveyance
114	8	3	Canal and inland navigation service
115	8	3	Boat and bargeman
116	8	3	Others connected with inland navigation
117	8	4	Shipowner
118	8	4	Seaman (merchant service)
119	8	4	Pilot
120	8	4	Others connected with sea navigation
121	8	5	Warehouseman
122	8	5	Others connected with storage
123	8	6	Messenger, porter (not Government), errand-boy
124	8	6	Others employed about messages
125	9	1	Land proprietor
126	9	1	Farmer
127	9	1	Grazier

128	9	1	Farmer's, grazier's wife
129	9	1	Farmer's, grazier's son, grandson, brother, nephew
130	9	1	Farmer's, grazier's daughter, granddaughter, sister, neice
131	9	1	Farm bailiff
132	9	1	Agricultural labourer (outdoor)
133	9	1	Shepherd
134	9	1	Farm servant (indoor)
135	9	1	Others connected with agriculture
136	9	2	Woodman
137	9	2	Others connected with arboriculture
138	9	3	Gardener
139	9	3	Nurseryman
140	9	3	Others connected with horticulture
141	10	1	Horse-dealer
142	10	1	Groom, horse-keeper, jockey
143	10	1	Farrier, veterinary surgeon
144	10	1	Cattle, sheep, dealer, salesman
145	10	1	Drover
146	10	1	Gamekeeper
147	10	1	Vermin-destroyer
148	10	1	Fisherman
149	10	1	Others engaged about animals
150	11	1	Bookseller, publisher
151	11	1	Bookbinder
152	11	1	Printer
153	11	1	Others engaged about publications
154	11	2	Actor
155	11	2	Others engaged about theatres
156	11	3	Musician (not teacher)
157	11	3	Musical instrument, -maker, dealer
158	11	3	Others connected with music
159	11	4	Engraver
160	11	4	Others employed about pictures and engravings
161	11	4	Others engaged about figures and carving
162	11	5	Artificial flower maker
163	11	6	Toy, -maker, dealer
164	11	6	Persons connected with shows, games and sports
165	11	7	Civil engineer
166	11	7	Pattern designer
167	11	7	Other designers and draughtsmen
168	11	8	Medallist and medal-maker
169	11	9	Watchmaker, clockmaker
170	11	9	Philosophical instrument maker
171	11	10	Gunsmith
172	11	10	Others engaged in manufacture of arms
173	11	11	Engine and machine maker
174	11	11	Tool-maker
175	11	11	Others dealing in tools and machines
176	11	12	Coachmaker
177	11	12	Others connected with carriage making
178	11	13	Saddler, harness-maker
179	11	13	Whip-maker
180	11	13	Other harness-makers
181	11	14	Shipwright, shipbuilder
182	11	14	Boat, barge, -builder
183	11	14	Others engaged in fitting ships
184	11	15	Surveyor
185	11	15	Builder
186	11	15	Carpenter, joiner
187	11	15	Bricklayer
188	11	15	Mason, pavior
189	11	15	Slater
190	11	15	Plasterer
191	11	15	Painter, plumber, glazier
192	11	15	Others engaged in house construction
193	11	16	Wheelwright

194	11	16	Millwright
195	11	16	Other implement makers
196	11	17	Dyer, scourer, calenderer
197	11	17	Others engaged in manufacture of chemicals
198	12	1	Cowkeeper, milkseller
199	12	1	Cheesemonger
200	12	1	Butcher, meat salesman
201	12	1	Butcher's wife
202	12	1	Provision curer
203	12	1	Poulterer, gamedealer
204	12	1	Fishmonger, dealer, seller
205	12	1	Others dealing in animal food
206	12	2	Soap-boiler
207	12	2	Tallow-chandler
208	12	2	Comb-maker (for manufactures)
209	12	2	Others dealing in grease and bones
210	12	3	Fellmonger
211	12	3	Skinner
212	12	3	Currier
213	12	3	Tanner
214	12	3	Other workers in leather
215	12	4	Feather, -dresser, dealer
216	12	5	Hair, bristle, -manufacture
217	12	5	Brush, broom, -maker
218	12	5	Other workers, dealers in hair
219	12	6	Woolstapler
220	12	6	Knitter
221	12	6	Woollen cloth manufacturer
222	12	6	Fuller
223	12	6	Worsted manufacturer
224	12	6	Stuff manufacturer
225	12	6	Clothier
226	12	6	Woollen draper
227	12	6	Carpet, rug, -manufacture
228	12	6	Other workers, dealers in wool
229	12	7	Silk manufacture
230	12	7	Silkmercer
231	12	7	Ribbon manufacture
232	12	7	Fancy goods manufacture
233	12	7	Embroiderer
234	12	7	Other workers dealing in silk
235	13	1	Greengrocer
236	13	1	Corn merchant
237	13	1	Miller
238	13	1	Flour-dealer
239	13	1	Baker
240	13	1	Confectioner
241	13	1	Others dealing in vegetable food
242	13	2	Maltster
243	13	2	Brewer
244	13	2	Licensed victualler, beershopkeeper
245	13	2	Licensed victualler, beershopkeeper's wife
246	13	2	Wine and spirit merchant
247	13	2	Sugar-refiner
248	13	2	Grocer
249	13	2	Tobacconist
250	13	2	Others dealing in drinks, stimulants
251	13	3	Oil and colourman
252	13	3	French-polisher
253	13	3	Other workers, dealers in oils, gums & c.
254	13	4	Timber merchant
255	13	4	Other dealers, workers in timber
256	13	5	Cork-cutter
257	13	5	Others dealing in bark
258	13	6	Sawyer
259	13	6	Lath-maker

260	13	6	Other wood workers
261	13	7	Cabinet-maker, upholsterer
262	13	7	Turner
263	13	7	Chair-maker
264	13	7	Box-maker
265	13	7	Others dealing in wood furniture
266	13	8	Cooper
267	13	8	Other makers of wood utensils
268	13	9	Frame-maker
269	13	9	Block and print cutter
270	13	9	Other wood tool makers
271	13	10	Basket-maker
272	13	10	Thatcher
273	13	10	Straw plait manufacture
274	13	10	Other workers in cane, rush, straw
275	13	11	Ropemaker
276	13	11	Sailcloth manufacture
277	13	11	Others working in hemp
278	13	12	Flax, linen, -manufacturer
279	13	12	Thread manufacture
280	13	12	Weaver (material not stated)
281	13	12	Draper
282	13	12	Lace manufacture
283	13	12	Cotton manufacture
284	13	12	Lint manufacture
285	13	12	Packer and presser (cotton)
286	13	12	Fustian manufacture
287	13	12	Muslin embroiderer
288	13	12	Calico, cotton, -printer
289	13	12	Calico, cotton, -dyer
290	13	12	Other workers, dealers in flax, cotton
291	13	13	Paper manufacture
292	13	13	Stationer
293	13	13	Paper-stainer
294	13	13	Paper-hanger
295	13	13	Other paper workers, dealers
296	14	1	Coal-miner
297	14	1	Coal, -merchant, dealer
298	14	1	Coal heaver or labourer
299	14	1	Chimney-sweeper
300	14	1	Gasworks service
301	14	1	Other dealers, workers in coal
302	14	2	Stone-quarrier
303	14	2	Slate-quarrier
304	14	2	Limestone, -quarrier, burner
305	14	2	Marble mason
306	14	2	Brick, -maker, dealer
307	14	2	Road labourer
308	14	2	Railway labourer
309	14	2	Other workers in stone, lime, clay
310	14	3	Earthenware manufacture
311	14	3	Earthenware and glass dealer
312	14	3	Tobacco-pipe maker
313	14	4	Glass manufacture
314	14	4	Other workers in glass
315	14	5	Salt, -agent, merchant, dealer
316	14	6	Water, -carrier, dealer
317	14	7	Workers, dealers in precious stones
318	14	8	Goldsmith, silversmith
319	14	8	Plater
320	14	8	Carver, gilder
321	14	8	Other workers in gold and silver
322	14	9	Copper-miner
323	14	9	Copper manufacture
324	14	9	Coppersmith
325	14	9	Other workers, dealers in copper

326	14	10	Tin-miner
327	14	10	Tinman, tin-worker, tinker
328	14	10	Other workers, dealers in tin
329	14	11	Zinc manufacture
330	14	11	Other workers, dealers in zinc
331	14	12	Lead-miner
332	14	12	Lead manufacture
333	14	12	Other workers, dealers in lead
334	14	13	Brass, -manufacture, founder, moulder
335	14	13	Locksmith, bellhanger
336	14	13	Brazier
337	14	13	White metal manufacture
338	14	13	Pin manufacture
339	14	13	Button-maker (all branches)
340	14	13	Wire, -maker, drawer
341	14	13	Wire, -worker, weaver
342	14	13	Other workers, dealers in mixed metals
343	14	14	Iron-miner
344	14	14	Iron, manufacture, moulder, founder
345	14	14	Whitesmith
346	14	14	Blacksmith
347	14	14	Nail manufacture
348	14	14	Anchorsmith, chainsmith
349	14	14	Boiler-maker
350	14	14	Ironmonger
351	14	14	File-maker
352	14	14	Cutler
353	14	14	Needle manufacture
354	14	14	Grinder (branch undefined)
355	14	14	Other workers, dealers in iron, steel
356	15	1	Labourer (branch undefined)
357	15	2	Mechanic, manufacturer, shopman, shopwoman
358	15	3	Others of indefinite occupations
359	16	1	Gentleman, gentlewoman, independent
360	16	1	Annuitant
361	16	1	Others of independent means
362	17	1	Dependent on relatives (not classed elsewhere)
363	17	1	Almsperson
364	17	1	Pauper of no stated occupation
365	17	1	Lunatic of no stated occupation
366	17	1	Others supported by the community
367	17	2	Prisoner of no stated occupation
368	17	3	Vagrants in barn, tents, etc.
369	18	1	Persons of no stated occupations or conditions